Reliability Engineering and System Safety 163 (2017) 69–78

Contents lists available at ScienceDirect

Reliability Engineering and System Safety

journal homepage: www.elsevier.com/locate/ress

Selection of risk reduction portfolios under interval-valued probabilities

Antti Toppila*, Ahti Salo

Systems Analysis Laboratory, Department of Mathematics and Systems Analysis, Aalto University School of Science, P.O. Box 11100, 00076 Aalto, Finland

ARTICLE INFO

Keywords:
Epistemic uncertainty
Interval-valued probabilities
Reliability allocation

ABSTRACT

A central problem in risk management is that of identifying the optimal combination (or portfolio) of improvements that enhance the reliability of the system most through reducing failure event probabilities, subject to the availability of resources. This optimal portfolio can be sensitive with regard to epistemic uncertainties about the failure events' probabilities. In this paper, we develop an optimization model to support the allocation of resources to improvements that mitigate risks in coherent systems in which interval-valued probabilities defined by lower and upper bounds are employed to capture epistemic uncertainties. Decision recommendations are based on portfolio dominance: a resource allocation portfolio is dominated if there exists another portfolio that improves system reliability (i) at least as much for all feasible failure probabilities and (ii) strictly more for some feasible probabilities. Based on non-dominated portfolios, recommendations about improvements to implement are derived by inspecting in how many non-dominated portfolios a given improvement is contained. We present an exact method for computing the non-dominated portfolios. We also present an approximate method that simplifies the reliability function using total order interactions so that larger problem instances can be solved with reasonable computational effort.

1. Introduction

The reliability of a system can be improved by using higher quality components, introducing redundant components, reducing the operational and environmental loads on the components, or improving the maintainability of the component, for instance [1]. The selection of components to be improved can be formalized as the *redundancy allocation problem*: Find which *portfolio* (i.e. combination) of improvements increases the reliability most, given that the portfolio must be feasible in that only some of the possible improvements can be implemented subject to relevant economical, technical or other constraints such as the available budget for improvements (for reviews, see [2–4]). The redundancy allocation problem has a fundamental role in the generation of decision recommendations and several efficient algorithms for solving it have been developed (see, e.g. [5]).

The classical formulation of the redundancy allocation problem assumes that the reliabilities before and after the improvements are known. However, in most practical instances there is epistemic uncertainty about the reliability [6]. This type of uncertainty – which is present in most redundancy allocation models – is also referred to as reducible, or non-stochastic uncertainty. It expresses the lack of knowledge about the probabilities experienced by the decision maker or the analyst (see also [7]). In reliability analysis, epistemic uncertainty has been modeled with methods such as Dempster–Shafer theory, Bayesian methods, fuzzy sets, possibility theory, and imprecise probability/reliability [8–15]. These methods have been used for modeling epistemic uncertainty in the redundancy allocation problem (e.g., [16–19]).

In this paper, we solve the redundancy allocation problem when the epistemic uncertainty about is characterized by interval valued probabilities. The uncertainty is characterized with lower and upper bounds on the probability that a component fails. These bounds determine an interval, within which this probability value can vary. The intervals for all components form a set of *feasible probabilities*.

When there is epistemic uncertainty about probabilities, the reliability allocation problem is no longer a standard optimization problem. In such cases, several methods have been developed to support the selection of a portfolio of improvements such that the epistemic uncertainty about the probabilities are properly taken into account. Depending on how the epistemic uncertainty is modeled, the computational methods for deriving the decision recommendations differ. For interval valued probabilities – which are used in this paper – such methods for modeling epistemic uncertainty in reliability allocation include the use of min-max robustness [20] and interval universal generator functions [17].

The model presented in this paper uses pairwise dominance to determine when one portfolio is better than another: A portfolio *dominates* another if it increases the reliability of the system at least

* Corresponding author.
E-mail addresses: atoppila@gmail.com (A. Toppila), ahti.salo@aalto.fi (A. Salo).

http://dx.doi.org/10.1016/j.ress.2017.02.005
Received 14 January 2016; Received in revised form 7 September 2016; Accepted 18 February 2017
Available online 21 February 2017
0951-8320/ © 2017 Elsevier Ltd. All rights reserved.

as much for all feasible probabilities and, in addition, increases the reliability of the system strictly more for some feasible probabilities. A decision maker should select a feasible portfolio which is non-dominated (ND), i.e., a feasible portfolio which is not dominated by any other feasible portfolio. Typically there are several ND portfolios, which are all reasonable candidates to be selected.

We build a framework analogous to the Robust Portfolio Modeling (RPM) framework by Liesiö et al. [21,22] to derive insights which aid in selecting the improvement portfolio. In this framework, the ND portfolios are used to derive decision recommendations that are robust with respect to epistemic uncertainty on which improvements should be included/excluded from any reasonable portfolio. Besides providing robust decision recommendations, another possible benefit of the framework is that several alternative portfolios (i.e. the ND portfolios) can indeed be justified (instead of a single portfolio). This, in turn, together with the robustness of the decision recommendations can strengthen the decision maker's confidence in the results of the model, especially when the decision is made jointly by several stakeholders. The method also helps guide the elicitation of more precise information by giving indications about which epistemic uncertainties should be reduced in order to obtain more specific decision recommendations [23].

The challenge in applying our framework is the computation of the ND portfolios. For this, we present an exponential time algorithm based on the implicit enumeration of all possible portfolios. We use the algorithm to solve an illustrative case that represents a residual heat removal system of a nuclear power plant. This example has 31 failure events, the probabilities of which could be reduced through improvements in the facility. We compare our results to the case where the selection of improvements is supported with an approximative heuristic based on the ranking of importance measures under interval valued probabilities [24].

In an earlier paper [24], we considered the components of a system in isolation using importance measures under interval valued probabilities. Also in this work, the decision recommendations are based on deriving a set of ND alternatives. To contrast our contribution with this work, we highlight the difference in the algorithmic development. When components n are considered in isolation, comparing all improvements to each other only requires n^2 dominance checks. However, in this paper, we consider a portfolio of improvements. Because a portfolio of n improvements can be selected in 2^n different ways, generating all portfolios and comparing them to each other for dominance is computationally intractable. Hence, this papers uses techniques and methods from exact portfolio optimization in the algorithmic development.

When the probabilities' intervals were very wide, our algorithm did not solve all non-dominated portfolios within reasonable time, when considering portfolios of more than five simultaneous improvements. For such situations, we use total order interactions introduced by [25,26] to simplify the reliability function of the system, and compute the non-dominated (ND) set with respect to it.

The rest of this paper is structured as follows. Section 2 formulates the optimal redundancy allocation problem as a portfolio optimization problem. Section 3 covers computational experiments related to optimal redundancy allocation and the analysis of epistemic uncertainties. Section 4 presents a method for simplifying the reliability function so that larger problem instances can be solved. Section 5 discusses and Section 6 concludes.

2. Methodological development for solving the optimal risk reduction portfolio problem

Consider a system consisting of n components, which fail with probability p_i, $i = 1, \ldots, n$. The reliability of the system is a function of the components' failure probabilities. More specifically, the system's reliability function is $r(p)$, where $p = (p_1, \ldots, p_n)$. It can be shown that this function is multilinear with respect to its arguments for both coherent and non-coherent systems (see, e.g., [26]). Furthermore, if the components' failures are statistically independent, we have

$$r(p) = \sum_{I \subseteq \{1,\ldots,n\}} \alpha_I \prod_{j \in I} p_j \tag{1}$$

for some coefficients $\alpha_I \in \mathbb{R}$ which are implied by the structure of the system, where $I \subseteq \{1, \ldots, n\}$ is a set valued index.

In typical applications, the multilinear function is sparse so that most of the coefficients α_I are zero. Thus from computational perspective, it is preferable to use a sparse data structure that saves only the non-zero coefficients and the indices of the variables that are included in the corresponding term.

Assume that the reliability of the system can be enhanced by improvement actions (referred to as *improvements*) which reduce the failure probability of a single component. We assume that each component can be improved once and improvement i changes the failure probability of component i from p_i to p_i^r for a cost c_i, i=1,...,n. We encode a portfolio $P \subseteq \{1, \ldots, n\}$ of improvements as the vector $z = (z_1, \ldots, z_n)$ such that

$$z_i = \begin{cases} 1 & \text{if } i \in P \\ 0 & \text{if } i \notin P. \end{cases}$$

Because there is a one-to-one mapping between the binary vector z and the corresponding set P, we also call z a portfolio. Under this notation, the set of all portfolios can be expressed as $\mathcal{Z} = \{0, 1\}^n$. The cost of a portfolio is $\sum_i c_i z_i$. A portfolio is feasible if its cost does not exceed a given budget B.

Given the improvement portfolio z, the probability of event i is $q_i = p_i(1 - z_i) + p_i^r z_i$ (note that if z_i=1, then $q_i = p_i^r$ and if z_i=0 then q_i=p_i). Consequently, the reliability of the system after applying the improvements in portfolio z is $r(z; p) = r(q)$, where $q = (q_1, \ldots, q_n)$. For instance, if the system consists of two components in parallel, then $r(p) = 1 - p_1p_2$ and $r(z; p) = r(q) = 1 - q_1q_2$ $=1 - (p_1^r z_1 + (1 - z_1)p_1)(p_2^r z_2 + (1 - z_2)p_2) = 1 - p_1p_2 + p_1p_2z_1 - p_1^r p_2 z_1 + p_1p_2z_2 - p_1p_2^r z_2 - p_1p_2z_1z_2 + p_1^r p_2 z_1z_2 + p_1p_2^r z_1z_2 - p_1^r p_2^r z_1z_2$.

The goal is to determine the portfolio z that reduces the risk – or equivalently increases the reliability of the system – as much as possible. More specifically, for a given p, the optimal portfolio is the solution to the problem

$$\max_z \quad r(z; p) \text{s.t.} \quad \sum_{i=1}^{m} c_i z_i \le B. \tag{2}$$

This is a knapsack optimization problem with a multilinear objective function and a linear budget constraint.

2.1. Interval-valued probabilities and dominance

As noted in Section 1, it is of interest to analyze how epistemic uncertainty about the probabilities captured by interval-values impacts the results of the analysis. Thus, assume that $p_i \in [\underline{p}_i, \overline{p}_i]$ for i=1,...,n, or equivalently $\underline{p} \le p \le \overline{p}$, where $\underline{p}_i$ and $\overline{p}_i$ are the lower and upper bounds of p_i. The set of feasible probabilities is $\mathcal{P}_F = \{p \in \mathbb{R}^n | \underline{p}_i \le p_i \le \overline{p}_i\}$. These bounds can be inferred, for instance, from the confidence intervals of the events' probability estimates, simulation [27–30], interval-probabilities [31], fuzzy probabilities [32], coherent lower and upper probabilities [7], imprecise reliability [12], and Dempster–Shafer theory [33,34], depending on what type of models are preferred by the analyst.

In our computational model, we assume that system reliability is monotone with respect to improvements, i.e. an improved component is always superior to an unimproved component. In line with this assumption, we model the impact of the improvements such that including improvement i to the portfolio reduces the failure probability of component i by a factor of a_i, $0 \le a_i \le 1$. Thus, the failure probability

of component i before the improvement is p_i and after the improvement is $p_i^r = a_i p_i \in [a_i \underline{p}_i, a_i \overline{p}_i]$. Here, a_i=0 represents a change that makes the failure of component i impossible, whereas $a_i = 1$ means that the failure probability of component i does not change.

With interval-valued probabilities, the solution to problem (2) is no longer unique, because with different values of p, the optimal portfolio may be different. However, some portfolios are inferior to others in that they are less reliable for *all* feasible p. For such settings, we define dominance as follows.

Definition 1. Portfolio z dominates portfolio z', denoted $z \succ z'$, iff $r(z; p) \geq r(z; p')$ for all $\underline{p} \leq p \leq \overline{p}$, and $r(z; p) > r(z'; p))$ for some $\underline{p} \leq p \leq \overline{p}$. The set of non-dominated portfolios is

$$\mathcal{Z}_N = \{z \in \mathcal{Z}_F | \nexists z' \in \mathcal{Z}_F : z' \succ z\},$$

where $\mathcal{Z}_F = \{z \in \mathcal{Z} | \sum_{i=1}^{m} c_i z_i \leq B\}$ is the set of feasible portfolios.A rational decision maker can limit her choices to non-dominated portfolios (see, e.g. [35,22,24]). This is because if she were to select a dominated portfolio, then it would be possible to select some non-dominated portfolio of improvements such that the system is as reliable for all feasible probabilities and more reliable for some feasible probabilities than this selected portfolio.

If the information about the probabilities becomes such that the lower bound of the probability is increased or upper bound is decreased (resulting in a narrower interval), then any portfolio that is dominated will stay dominated, and the new set of non-dominated portfolios will be a subset of the current non-dominated set.

The set of non-dominated portfolios can provide guidance on which improvements should be included in a portfolio through the use of core indices [35], which are defined as follows.

Definition 2. The core index CI of improvement i is defined as

$$CI_i = \frac{|\{z \in \mathcal{Z}_N | z_i = 1\}|}{|\mathcal{Z}_N|},$$

where $|\cdot|$ denotes the number of elements in a set.

The core indices divide improvements into three categories, namely core (CI_i=1), exterior (CI_i=0) and borderline improvements ($0 < CI_i < 1$). Core improvements are in all non-dominated portfolios, and hence a rational decision maker will choose these improvements in her portfolio. She will not select an exterior improvement into her portfolio, because doing so would also make her portfolio dominated. Borderline improvements could be included or excluded from the portfolio depending on what other improvements are in the portfolio without necessarily resulting in a dominated or non-dominated portfolio.

The core index can be used to guide the elicitation of further information. Assume that the elicitation leads to assigning the probabilities tighter intervals that are contained in the original intervals. Then eliciting information about core and exterior projects cannot lead to that a ND portfolio would be ND under the new tighter intervals (see [35]). Hence, eliciting information about them is not useful. Instead, one should elicit information about borderline projects only.

The CI can also be used to guide a sequential decision making process of selecting improvements. Clearly, all core improvements should be selected and exterior improvements should not be selected. To quickly identify a single portfolio or a smaller subset of portfolios, one should consider the case where an improvement with small core index is selected or an improvement with a large core index is not selected. The ND portfolios that are compatible with this restriction is then relatively low, which may help the decision maker in analyzing the remaining portfolios. Moreover, the CI has also be given a probabilistic interpretation [23]: If it is equiprobable that every ND portfolio is the best one, and one maximizes the expected number of "correct" actions in the portfolio, then a heuristic for this maximization could be to select the portfolio which has highest sum of actions' CI. In group decision making, the CI can be used measuring how well the group members agree about decision recommendations [36].

2.2. Computation of dominance

In our computational development, we build upon the framework by [24] in which dominance is evaluated by examining the non-negativity of a multilinear function. Specifically, in order to check if portfolio z dominates portfolio z', the first condition in Definition 1 is $r(z, p) \geq r(z', p)\ \forall\ p \in \mathcal{P}_F$, which is equivalent to the non-negativity of the multilinear function $g(p) = r(z, p) - r(z', p)$. The second condition of dominance follows from the first condition if all intervals are non-singleton sets (Proposition 1 in [24]). If some of the intervals are singleton sets (i.e. point estimates), then these probabilities would have unique values that can be substituted into $g(p)$. This substitution results in a new multilinear function g', which contains only variables that are constrained in non-singleton sets. Thus dominance can be characterized by the non-negativity of a multilinear function. The non-negativity of a multilinear function can be checked with the algorithm by [24], for instance.

2.3. Computation of non-dominated portfolios

The set of all non-dominated portfolios can in principle be computed by enumerating all the possible portfolios and by performing pairwise dominance checks. However, the number of portfolios increases exponentially with the number of possible improvements, and thus this approach requires too much computational resources. Therefore this problem is best approached with portfolio optimization algorithms such as implicit enumeration and dynamic programming, which have been used for solving the redundancy allocation problem (see references in [3,2]).

The reliability function $r(p)$ of coherent systems is monotone in its arguments. Without loss of generality, we assume that $r(p)$ is decreasing in its arguments. We also assume that $r(z; p)$ is increasing in z, because no decision maker would select an alternative that reduces the reliability.

The basis for the implicit enumeration algorithm is in Table 1. It consists of two distinct phases. The first phase on lines 1–22 is the implicit enumeration phase which yields a superset of the non-dominated set. The second phase on line 23 removes dominated portfolios from the superset given by the first phase such that only non-dominated portfolios remain. The details of the algorithm are as follows.

On line 1, the current solution z, and the active alternative k are initialized. Lines 4–8 and 16–22 of the algorithm enumerates all possible portfolios using a last-in-first out ordering (see, e.g. [37,38]). On line 9, the feasibility of the portfolio is checked; if this is the case, then on line 10, the portfolio is added to the set of potential non-dominated portfolios Z. On line 11, the lowest reliability that the portfolio can have is computed and compared to the reliability of the so called minmax portfolio, which is the portfolio with the lowest reliability lower bound of the portfolios found so far. If the lowest bound is higher, then the current portfolio is updated to this position.

Lines 13–15 form the bounding phase involving two types of bounds. The first type of bounding is based on feasibility. That is, if at an iteration in the `Forward loop` for some k we have that $\sum_{i=1}^{k} c_i z_i \geq B$, then every further iteration made in the `Forward loop` corresponds to an infeasible solution until a Back-track step is made such that $z_k \leftarrow 0$. Thus, it not necessary to explicitly account for these infeasible portfolios. Instead, the `Forward-loop` can be terminated prematurely and the `Backtrack-step` can be taken instead.

The second bound is based on strict dominance: Assume that for a given k, portfolio z^* strictly dominates every feasible portfolio which is of the form $(z_1, \ldots, z_k, x_{k+1}, \ldots, x_m)$ where $x_j \in \{0, 1\}$. Then the `Forward loop` will yield only portfolios that are dominated until a `Backtrack-step` $z_k \leftarrow 0$ is made. Thus, it is possible to do this `Backtrack-step` immediately without losing any non-dominated portfolios. Note that z^* is initialized as the empty set. Hence, we define that $r(\varnothing, \underline{p}) = -\infty$ for

Table 1
Algorithm: NonDominatedSetDepthFirst.

Input Monotonic dominance relation $\succ$ over portfolios $z \in \mathcal{Z}_F \subseteq \{0, 1\}^m$.
Output Set of all non-dominated portfolios ($\mathcal{Z}_N$).

1: $z \leftarrow (0, \ldots, 0)$, $z^* = \emptyset$, $Z \leftarrow \{\}$, $k \leftarrow 0$
2: **if** $\sum_i c_i z_i \leq B$ **then**
3: $Z \leftarrow \{z\}$
4: **repeat**
5: `Forward-loop`
6: **repeat**
7: $k \leftarrow k + 1$
8: $z_k \leftarrow 1$
9: **if** $\sum_i c_i z_i \leq B$ **then**
10: $Z \leftarrow Z \cup \{z\}$
11: **if** $r(z; \underline{p}) > r(z^*; \underline{p})$ **then**
12: $z^* \leftarrow z$
13: `Bounding phase`
14: **if** $\sum_i c_i z_i \geq B$**or**

$z^* \succ (z_1, \ldots, z_k, x_{k+1}, \ldots, x_m)\ \forall\ p \in \mathcal{P}_F, (z_1, \ldots, z_k, x_{k+1}, \ldots, x_m) \in \mathcal{Z}_F$

then
15: **break loop**
16: **until** k=m
17: `Backtrack-step`
18: $z_m \leftarrow 0$
19: $k \leftarrow \max(\{j \in \{1, \ldots, m\}: z_j = 1\} \cup \{0\})$
20: **if** $k > 0$ **then**
21: $z_k \leftarrow 0$
22: **until** k=0
23: Remove from Z dominated solutions by pairwise comparison.

cases where the starting portfolio $(0, \ldots, 0)$ is infeasible.

Before executing line 23, the first part of the algorithm yields the set Z that is guaranteed to contain all non-dominated portfolios. In the second phase, pairwise comparisons of portfolios in Z are carried out to remove those dominated portfolios that were not screened in the bounding phase. After this second phase, only non-dominated portfolios remain.

The above steps are straightforward to implement except the second bounding criterion on line 14; we discuss this in more detail in the following section.

2.4. Computation of bounds

The second bounding criterion in the algorithm in Table 1 is

$$r(z^*; p) > r(z_1, \ldots, z_k, x_{k+1}, \ldots, x_m; p)\ \forall\ p \in \mathcal{P}_F, (z_1, \ldots, z_k, x_{k+1}, \ldots, x_m) \in \mathcal{Z}_F.$$

Checking this inequality is a combinatorial problem that can be solved by checking $r(z, p) - r(z', p) \geq 0$ for all $p \in \mathcal{P}_F$ and $z' = (z_1, \ldots, z_k, x_{k+1}, \ldots, x_m) \in \mathcal{Z}_F$ as described in the previous section. However, this makes it necessary to evaluate 2^{m-k-1} portfolios, and therefore an exact evaluation as a part of an already exponential time algorithm is not computationally viable. Instead, we derive a computationally inexpensive condition that is sufficient (but not necessary) for the second condition to hold.

Consider the function

$$g(x; p) = g(x_{k+1}, \ldots, x_m; p) = r(z^*; p) - r(z_1, \ldots, z_k, x_{k+1}, \ldots, x_m; p).$$

This function is multilinear in p and x. It is decreasing in x, because improving the reliability of a component will improve the reliability of a coherent system. Because it is multilinear, the function $g(x; p)$ can be factored in the form:

$$g(x; p) = a(x_{-i}; p)x_i + b(x_{-i}; p),$$

where

$$a(x_{-i}; p) = \sum_{\{I \subseteq \{1,\ldots,m\} | i \in I\}} a_I(p) \prod_{j \in I \setminus \{i\}} x_j$$

and

$$b(x_{-i}; p) = \sum_{\{I \subseteq \{1,\ldots,m\} | i \notin I\}} a_I(p) \prod_{j \in I} x_j$$

are multilinear functions and $x_{-i} = (x_{k+1}, \ldots, x_{i-1}, x_{i+1}, \ldots, x_m)$, $i = k + 1, \ldots, m$. This form shows that the change from x_i=0 to x_i=1 changes the function value by $a(x_{-i}; p)$.

We next derive a lower bound on $a(x_{-i}; p)$. Any coherent system can be described through a sequence of subsystems (fault tree) including only AND and OR gates. For these gates, it is straightforward to show that the total increase in system reliability achieved through joint improvements in the reliability of two or more of the components is less than the sum of reliability improvements attained by improvements of these individual components. Hence, by induction, this property holds for any coherent system. This implies that the coefficient $a(x_{-i}; p)$ is minimal when $x_{-i} = (0, \ldots, 0)$. Thus, it is of interest to establish a lower bound for $a(0, \ldots, 0; p) = a_\emptyset^{-i}(p)$. Because $a_\emptyset^{-i}(p)$ is multilinear with respect to p, it follows that $a_\emptyset^{-i}(p) = \sum_{I \subseteq \{1,\ldots,n\}} a_I^{-i} \prod_{j \in I} p_j$ for some unique coefficients a_I^{-i}. A lower bound on $a_\emptyset^{-i}(p)$ can be computed as

$$\underline{a}_\emptyset^{-i} := a_\emptyset^{-i} + a_+^{-i}(\overline{p}) + a_-^{-i}(\underline{p}), \quad (3)$$

where $a_\emptyset^{-i}$ is the constant, $a_+^{-i}(p)$ is the sum of the terms with positive

coefficients a_I^{-i} and $a_-^{-i}(p)$ is the sum of the terms with negative coefficients a_I^{-i}.

Typically not all of the x_i can be assigned a value of 1 because of the budget constraint. To obtain a better bound, we use a continuous knapsack problem for bounding the lowest value that the simultaneous change of multiple x_i:s can have. The remaining budget is given by $B - \sum_{i=1}^{k} c_i z_i$. We seek to show that regardless of how the x_i, $i = k+1, \ldots, n$ are chosen, the inequality $g(x; p) \geq 0$ holds. Now, consider the following continuous knapsack problem

$$\min_x \sum_{i=k+1}^{m} \underline{a}_{\varnothing}^{-i} x_i \text{subject to} \sum_{i=k+1}^{m} c_i x_i \leq B - \sum_{i=1}^{k} c_i z_i \quad 0 \leq x_i \leq 1,$$

$i = k+1, \ldots, m.$

This is a standard continuous knapsack optimization problem which be solved efficiently (for details, [38]). The optimal value $\tau \leq 0$ of this problem gives a bound on how much at most the value of $g(x, p)$ can be reduced when changing x_i from 0 to 1 under a budget constraint. Thus $r(z^*; p) - r(z'; p) \geq r(z^*; p) - r(z; p) + \tau$. We summarize the implications of this discussion in the following lemma.

Lemma 1. *Consider portfolios z^* and z such that $B - \sum_{i=1}^{k} c_i z_i \geq 0$ and $z_{k+1}, \ldots, z_m = 0$. Let $\tau \leq 0$ denote the optimal value of the problem*

$$\min_x \sum_{i=k+1}^{m} \underline{a}_{\varnothing}^{-i} x_i \text{subject to} \sum_{i=k+1}^{m} c_i x_i \leq B - \sum_{i=1}^{k} c_i z_i 0 \leq x_i \leq 1,$$

$i = k+1, \ldots, m,$

where $\underline{a}_{\varnothing}^{-i}$ are given by Eq. (3). If $r(z^*; p) - r(z; p) + \tau \geq 0 \; \forall \, p \in \mathcal{P}_F$, then

$r(z^*; p) > r(z_1, \ldots, z_k, x_{k+1}, \ldots, x_m; p) \; \forall \, p \in \mathcal{P}_F$ *and* $(z_1, \ldots, z_k, x_{k+1}, \ldots, x_m) \in \mathcal{Z}_F$.

Proof. See the discussion above.□

Let us relate the derived bound to previous work. Consider the function

$$g(x) = \sum_{I \subseteq \{1,\ldots,m\}} \alpha_I \prod_{j \in I} x_j$$

and the problem

$$\min_x \quad g(x) \tag{4}$$

$$\text{subject to} \quad \sum_i c_i x_i \leq B \tag{5}$$

$$0 \leq x_i \leq 1. \tag{6}$$

We first prove a property of the optimal solution that was shown by [39] for symmetric multilinear functions. By factoring with respect to x_i, we get $g(x) = a(x_{-i})x_i + b(x_{-i})$, where $a(x_{-i}) = \sum_{I \subseteq \{1,\ldots,m\}: i \in I} \alpha_I \prod_{j \in I \setminus \{i\}} x_j$ and $b(x_{-i}) = \sum_{I \subseteq \{1,\ldots,m\}: i \notin I} \alpha_I \prod_{j \in I} x_j$ are multilinear functions. Using this factoring, the objective function can be written as

$$g(x) = \frac{1}{m} \sum_i g(x) = \frac{1}{m} \left(\sum_i a(x_{-i}) x_i + \sum_i b(x_{-i}) \right).$$

Assume that the optimal solution to the problem above is x^*. Consider the following related problem:

$$\min_x \quad \frac{1}{m} \left(\sum_i a(x_{-i}^*) x_i + \sum_i b(x_{-i}^*) \right) \text{subject to} \quad \sum_i c_i x_i \leq B 0 \leq x_i \leq 1. \tag{7}$$

Clearly, $x = x^*$ is a feasible solution of this optimization problem whose value at x^* is the same as that of the original problem (4) and (5); thus, (7) is a relaxation of the original problem (4)–(6). Next, we prove that the solutions are equivalent: The necessary and sufficient Karush–Kuhn–Tucker (KKT) condition for x being a local optimal solution of the former and latter problem is

$$\frac{\partial f}{\partial x_i} + u_0 \frac{\partial}{\partial x_i} \left(\sum_i c_i x_i - B \right) + \sum_i u_i (x_i - 1) + \sum_i u_{m+i} (-x_i) = 0 \quad \forall \, i$$

for some $u = (u_0, \ldots, u_{2m}) \leq 0$. For both problems, $\frac{\partial f}{\partial x_i} = a(x_{-i})$ and thus the KKT conditions are identical. Therefore x^* (which is the global optimal solution of the original problem) is a local optimal solution of the latter problem. Problem (7) is a linear programming problem, which implies that a local optimum is a global optimum as well. Therefore, x^* is its global optimal solution.

For optimizing the latter problem, the multiplication by $1/m$ and substraction of the constant $\sum_i b(x_{-i}^*)$ can be disregarded. Therefore the optimal solution to the original problem is the same as the solution of the continuous knapsack problem

$$\max_x \quad \sum_i a(x_{-i}^*) x_i \text{subject to} \quad \sum_i c_i x_i \leq B 0 \leq x_i \leq 1. \tag{8}$$

The continuous knapsack problem can be solved as follows (see, e.g. Section 2 in [38]): Without loss of generality, assume that $a(x_{-i}^*) > 0$, $c_i > 0$, $B > 0$, $\sum_i c_i > B$, $c_i \leq B$ and that the variables are sorted such that $a(x_{-i}^*)/c_i \geq a(x_{-(i+1)}^*)/c_{i+1}$ for all i. If these assumptions do not hold, then the problem can be transformed using simple operations to an equivalent knapsack problem where these assumptions hold, or the solution (or the solution with regard to some variables) is trivial.

The critical item x_j is defined by the index $j = \min\{k: \sum_{i=1}^{k} c_i > B\}$. Then the continuous knapsack problem has an optimal solution given by x_i=1 for $i < j$, x_i=0 for $i > j$ and $x_j = B - \sum_{i<j} c_i x_i$. The solution of the continuous knapsack problem is unique if the set $J = \{i: a(x_{-i}^*)/c_i = a(x_{-j}^*)/c_j\}$ satisfies $J = \{j\}$, that is, there are no other variables that has the same benefit to cost ratio as the critical item. Otherwise, the solution is not unique and all possible solutions are given by x_i=1 for $i \in \{1, \ldots, j-1\} \setminus J$, x_i=0 for $i \in \{j, \ldots, m\} \setminus J$ and $x_i \geq 0$, $\sum_{i \in J} c_i x_i = B - \sum_{k=1}^{\max(\{1,\ldots,j-1\} \setminus J)} c_k x_k$ for $i \in J$.

The solution of the original problem is unique if problem (8) (and hence also problem (7)) has a unique solution and consists of at most one variable that is not 0 or 1.

The above result characterizes the optimal solution in way that resembles Theorem 2(i) and its Corollary 2.1 by [39] for symmetric multilinear functions. The results of [39] was used by [37] to bound the value of a symmetric multilinear function in an implicit enumeration algorithm, in way similar to our method.

Problem (8) helps establish bounds for the branch-and-bound algorithm. Although we do not know $a(x_i^*)$ before solving the problem, we are able to find easily computable bound on these coefficients. This bounding is based on replacing $a(x_i^*)$ in (8) with $\overline{a}_i \geq a(x_i^*)$. Hence, the problem of this new solution will give an upper bound of the optimal value of the knapsack problem. This optimal value of the knapsack problem will then give a bound on how much the objective function can increase. Thus evaluating $g(x)$ at x=0, gives the most conservative bound.

In some instances, the proposed bound may be weak and its efficiency depends on how the function is represented. In general, the bound is better, the smaller the values of the nonlinear terms of $g(p)$ are. In particular, this bound can be expected to work well if p_i are close to zero (which is often the case in reliability analysis).

3. Computations and results

This section presents numerical computations which were carried out using a standard laptop with 8 GB RAM, 64-bit Windows 7, and Intel(R) Core(TM) i5-4300U CPU @ 1.90 GHz processor with two cores. We did not use parallelization techniques, and consequently only a single core was used.

In our computational analysis, we consider the same residual heat removal system of a nuclear power plant as [24]. In this system, there

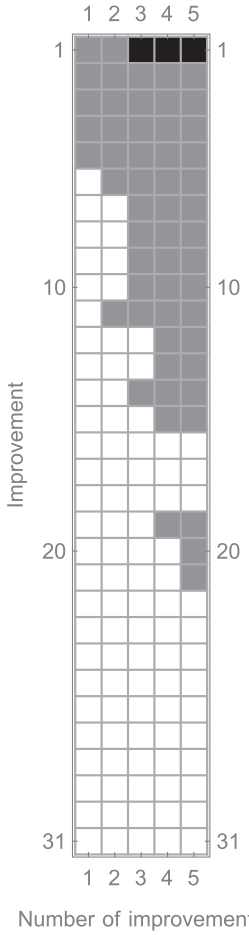


Fig. 1. Core indices in the residual heat removal system example.

are 31 failure events whose probabilities are within the lower and upper bounds defined by 90% confidence intervals. The system is characterized by 147 minimal cut sets which contain at most three events each (for details about the minimal cut sets and interval-probabilities, see [24]). The system can be improved by eliminating the event, that is reducing the probability of component failure zero. The portfolio of improvements is constrained by how many events can be eliminated. In the modeling framework presented in Section 3, these are modeled by setting a_i=0, c_i=1, and B is the number improvement that can be taken simultaneously.

We solved the non-dominated portfolios for this problem for portfolios that included 1–5 simultaneous improvements. The core indices based on these non-dominated sets are shown in Fig. 1. In this figure, the core indices are coded: Black indicates that the component is a core component, white indicates that the component has core index 0 (exterior component), and gray indicates that the component has a core index between 0 and 1 (borderline component). The algorithm did not terminate in reasonable time in cases where the optimal portfolio included 6 or more simultaneous improvements. The explanation can be seen from Table 2: The first part of the algorithm, which implicitly enumerates all portfolios, generated almost 15 000 non-dominated portfolios in about an hour. To remove dominated portfolios, second part of the algorithm removes dominated portfolios by pairwise dominance checks, which in this case yields in worst case about $(15\,000^2 - 15\,000)/2 = 112\,492\,500$ comparisons.

From Fig. 1, it can be seen that when only one improvement can be included, then only improvements 1–5 have a non-zero core index and they are all borderline, indicating that any of them could be selected. When the at most two improvements can be selected, then also improvements 6 and 11 become borderline. When three to five

Table 2
Computations of the residual heat removal system.

Number of improvements in portfolio	1	2	3	4	5	6
Number of feasible portfolios	31	465	4495	31 465	169 911	736 281
Number of potential non-dominated portfolios	5	70	242	2265	6764	14 609
Number of minmax potential ND portfolios	5	43	173	1592	5794	–
Number of non-dominated portfolios	5	9	21	58	133	NA

improvements can be selected, then more improvements become borderline. The most significant change is that improvement 1 becomes core. Thus as a summary, if only a few improvements can be selected, then no other improvements become core, which suggests that improvement 1 should have the highest priority regardless of which other improvements are selected.

Next we compare our analysis to the analysis in [24], where recommendations are derived based on Fussell–Vesely (FV) importance measure. In that paper, improvements with regard to components 1–5 are identified as non-dominated with respect to all other improvements (to avoid confusion, we abbreviate this as pairwise or pw-ND). When a single improvement can be selected to the portfolio, our method identifies this same set of improvements to be borderline and all others as exterior. However, when more improvements can be added to the portfolio, we see that more improvements become viable: when five improvements can be selected, a total of 15 improvements are borderline or core, compared to the risk measure based analysis, where only improvements 1–5 are pw-ND. We also see that for budgets 3–5, improvement 1 is a core improvement. In comparison, the FV measure based analysis does not depend on how many improvements are chosen and hence the decision recommendation does not change either. Therefore the core status of improvement 1 for large budgets is not detected.

We also compute the pw-ND components with respect to the Birnbaum (B) measure using the method by [24]. Then the pw-ND improvements are 1, 6, 14, 20, and 31. This ranking is quite different from the ranking based on the FV measure (above), because the B measure does not depend on the probability of the components' failure. Therefore the importance of events that have low probability tends to be exaggerated. In this case, we see that the B measure detects the importance of component 20, which belongs to some ND portfolios at budget level 5. However, as is evident from the figure, also other components than component 20 could be in the ND portfolio.

To solve the ND set with a larger budget, we reduced uncertainties by shrinking the interval endpoints towards the interval mean values so that the interval width was halved. As a result, there are fewer non-dominated portfolios and the range of values that a portfolio can have will become narrower so that the problem becomes easier to solve. With this setting, we could solve the problem to a budget level of 15, corresponding to about 50% of the events. The results are in Fig. 2, where we see the impact of multiple improvements. At low budget levels, the top three events are borderline or core events (as before in the case with full width intervals, event 1 becomes core at budget level 2 and above). Event 2 becomes core for budgets 3–6, but loses this status for budgets 7–12. This affect is attributed to the non-linearity of the reliability function, because in a standard knapsack problem, events are taken in the order of highest benefit to cost. Thus, once an improvement becomes core, this status should not be lost. An other special case is event 11, which becomes borderline at budget 3 and is the fourth event to become core when the budget is increased (Table 3).

We also sought to narrow down the number of events that can be included into the portfolio, while keeping the intervals at full width. This effectively reduces the dimension of the search space, but the uncertainties about the probabilities are fully propagated. Thus con-

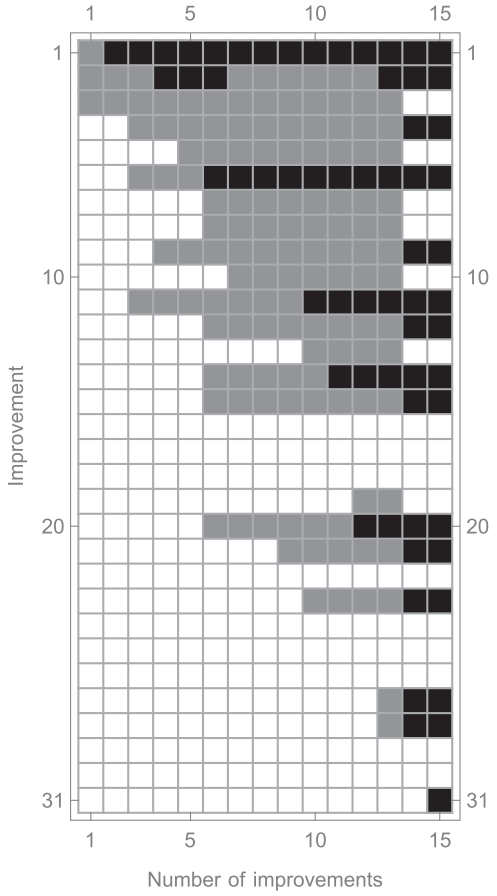


Fig. 2. Core indices in the residual heat removal system example with narrower intervals.

sider that events 1–10 form a short list from which a portfolio for improvements are selected. All events 1–31 have interval uncertainty about them, but the events that are not on the shortlist cannot be selected for improvement. These results are in Table 4. Here we see that the number of non-dominated portfolios is significantly reduced. Also the number of portfolios found by the implicit enumeration algorithm are very low, which explains why the pairwise comparisons for removing dominated events do not impede solving the problem.

4. Simplifying the reliability function by removing insignificant interactions

Importance measures can be used as a heuristic for solving the redundancy allocation problem (see, e.g. [40,41]). Then, by sequentially selecting components of the system for improvement and computing importance measures, one can find good solutions for reliability allocation problems. For instance the Birnbaum importance measure, which is defined as the probability that a basic event i becomes critical to the system, and which can be computed by

$$B_i(p) = \frac{\partial r(p)}{\partial p_i},$$

can be used for this purpose. From an optimization point of view, this measure linearizes the change with respect to each probability and recommends improving the system in the direct of the steepest accent. This can be modified to take into account the amount of improvement that can be made, the rate of change can be multiplied by the change in that probability, that is $B(p)|p_i^r - p_i| = B(p)\Delta p_i$.

The downside of using the Birnbaum measure for a optimization heuristics is that it does not account for interactions. To overcome this limitation, we use the method of [25]. In this method, the change of the reliability is considered in *sensitivity cases*: These cases consider simultaneous change in event probabilities such that $p_i \to p_i + \Delta p_i \ \forall\ i$, and the change is decomposed into three types of interactions measures (Proposition 7, [25]) where the measures are given by the equations

$$\beta_i^1 = B_i(p)\Delta p \tag{9}$$

$$\beta_{i_1,\ldots,i_n}^k = J_{i_1,\ldots,i_n}^k \Delta p_{i_1} \cdots \Delta p_{i_k} \tag{10}$$

$$\beta_i^T = B_i(p)\Delta p_i + \sum_{k=2}^{T} \sum_{i<i_2<\cdots<i_k} J_{i,i_2,\ldots,i_k}^k \cdot \Delta p_{i_1} \cdots \Delta p_{i_k}, \tag{11}$$

where

$$J_{i_1,\ldots,i_n}^k = \frac{\partial^k r(p)}{\partial p_{i_1} \cdots \partial p_{i_k}}$$

is the Joint Reliability Importance [see e.g. 40, p. 113]. The first components are the linearized changes. The second components are the changes caused by interactions. The third is the Total Order Importance (TOI) measure, as introduced by [26] (for further development, see [40], pp. 136–137). Note that in this paper, we use the unscaled versions of these measures. The scaled versions can be computed by dividing the measures with the total change related to the sensitivity case.

As noted in [25], Eqs. (9)–(11) can be evaluated by computing the reliability function $r(p)$ at $2n+2$ different points, making the computation of these measure efficient. These measures can be used for decomposing the change of a single probability to its individual impact and to an interaction impact. Here, β_i^1 is a measure of the individual impact of a variable and

$$\beta_i^I := \beta_i^T - \beta_i^1 \tag{12}$$

is a measure of the interaction with other variables.

As discussed in [26,25], these measures serve as an opportunity to find the most important interactions that drive the changes in reliability with respect to changes in event probabilities. Notably, if the changes are small, interactions do not play any role at all ([25], Proposition 5).

4.1. Elimination of insignificant interactions and variables

The elimination procedure proposed by [25] is as follows

Table 4
Computations of the residual heat removal system with short list 10.

# Improvements	1	2	3	4	5	6	7	8	9	10
#pot. ND portfolios	5	23	35	81	112	94	60	26	8	1
#ND portfolios	5	8	14	27	39	35	21	10	4	1

Table 3
Computations of the residual heat removal system with narrower intervals.

# Improvements	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
#full & feasible portf.	31	465	4.5k	31.5k	169.9k	0.7M	2.6M	7.9M	20.2M	44.6M	84.7M	141M	206M	265M	300M
#pot. ND portfolios	3	3	5	32	169	80	99	203	344	280	122	115	116	259	205
#ND portfolios	3	2	5	7	8	12	24	33	29	23	18	16	18	1	1

- Estimate the quantities $\beta_i^T/\Delta f$, $\beta_i^I/\Delta f$ and $\beta_i^1/\Delta f$, where Δf is the total change of the sensitivity case for i=1,...,n
- See if $\beta_i^I/\Delta f \ll \beta_i^1/\Delta f$. If this holds, then individual effects are the most significant for this variable and interactions can be neglected. An automated procedure could use the quantity

$$\epsilon_i = \frac{|\beta_i^I|}{|\beta_i^T|}, \quad i = 1, \ldots, n$$

as a decision criterion. Thus if this quantity is below a given threshold, say 10%, then the interaction could be considered to be negligible, because it would explain only a minor part of the change in that variable.

We build on this framework to simplify the reliability function so that it would be suitable for optimization. To achieve this, we establish guidelines for simplification of the reliability function with respect to those interactions that can be considered negligible.

Consider the function $r(z, p) = r(q)$, where $q_i = p_i(1 - z_i) + a_i p_i z_i$, where $0 \le a_i \le 1$, as a function of the variable vector $(z, p) = (z_1, \ldots, z_n, p_1, \ldots, p_n)$. For the purposes of dominance, it is of interest to know how the function changes with respect to the function varying between its lower and upper bounds. Thus for p, it is interesting to compute a sensitivity case where the values of p are changed from $\underline{p}$ to $\overline{p}$.

Next we discuss what sensitivity case should be considered for z. Clearly, changing every z_j variable from 0 to 1 is not that interesting, if a budget constraint would require that only a small subset of the variables can be set to 1. Therefore, we propose that in the sensitivity case, where variables with indices $\ell \in L \subset \{1, \ldots, n\}$ such that $\sum_{\ell \in L} c_i \le B$ are allowed a change from 0 to 1. Here the set L can be determined by taking variables in the order of a prior belief about which events should be improved until the budget is exhausted. Such an order can be derived using importance measures, such as Birnabaum, Fussell-Vesely, or Risk Achievement Worth, for instance (for details, see e.g. [40]).

The total order interactions β_i^I, the individual impacts β_i^1 and interaction impacts β_i^I should be computed based on the sensitivity case presented in the previous paragraph. Then, as proposed in the original method, variables are ordered from the most to the least relevant according to $|\beta_i^T|$. Then, a threshold rule such as the one presented above can be used for determining which variables should be simplified either by linearizing their impact or by substituting them with constants.

4.2. Guidelines for simplification

Here, we develop guidelines for simplification that can be done if an interaction is considered insignificant. Consider the following rules:

1. If β_i^T and β_i^I are small, then fix this event's probability to the interval upper bound.
2. If β_i^T and β_i^1 are large, and β_i^I is small, then fix the value of the probability of all other events to their interval upper bounds in all terms which the event is present.
3. If β_i^T is small but β_i^I and β_i^1 are large, then give high priority to this variable in branching.

The rationale for these rules is as follows: The first rule is justified because if the total impact is small and it does not interact significantly either, then this variable can be treated as a constant. The second rule says that if the interactions are insignificant but the individual impact is significant, then it is better to treat changes on this variable as a linear function, without allowing the probability to interact with other probabilities. The third rule is motivated by the fact that the bounds proposed to used in the implicit enumeration algorithm and the dominance checking algorithm are based on linearization. Branching on such a variable removes the nonlinearity associated with that variable, which potentially accelerates the termination these algorithms.

An important note about these rules is that it does not matter in which order they are employed. Thus even if they are applied repeatedly and interactively, the end result will be the same.

Table 5
Total order interactions of the RHRS case.

Variable index	β_i^T	β_i^1	β_i^I
1	4.42113×10^{-1}	4.42113×10^{-1}	4.21885×10^{-15}
2	4.61616×10^{-2}	3.57635×10^{-1}	-3.11473×10^{-1}
3	4.11311×10^{-2}	3.12343×10^{-1}	-2.71212×10^{-1}
4	6.14858×10^{-3}	5.65826×10^{-2}	-5.0434×10^{-2}
5	5.21867×10^{-3}	4.82103×10^{-2}	-4.29916×10^{-2}
6	2.12518×10^{-2}	2.12518×10^{-2}	0.
7	3.96072×10^{-3}	3.0077×10^{-2}	-2.61163×10^{-2}
8	3.96072×10^{-3}	3.0077×10^{-2}	-2.61163×10^{-2}
9	2.22785×10^{-3}	1.72601×10^{-2}	-1.50323×10^{-2}
10	1.98507×10^{-3}	1.50743×10^{-2}	-1.30892×10^{-2}
11	1.32923×10^{-3}	3.28293×10^{-2}	-3.15×10^{-2}
12	7.96104×10^{-4}	6.16778×10^{-3}	-5.37167×10^{-3}
13	7.09349×10^{-4}	5.38668×10^{-3}	-4.67733×10^{-3}
14	2.42877×10^{-3}	2.42877×10^{-3}	-4.21321×10^{-15}
15	6.20899×10^{-4}	4.81038×10^{-3}	-4.18948×10^{-3}
16	7.59309×10^{-5}	1.48437×10^{-2}	-1.47678×10^{-2}
17	7.23678×10^{-5}	1.40417×10^{-2}	-1.39693×10^{-2}
18	3.12146×10^{-4}	7.80365×10^{-3}	-7.4915×10^{-3}
19	5.53237×10^{-4}	4.20119×10^{-3}	-3.64796×10^{-3}
20	1.51798×10^{-3}	1.51798×10^{-3}	-4.21321×10^{-15}
21	2.54878×10^{-4}	1.97465×10^{-3}	-1.71977×10^{-3}
22	2.27102×10^{-4}	1.72458×10^{-3}	-1.49748×10^{-3}
23	1.59221×10^{-4}	1.23356×10^{-3}	-1.07433×10^{-3}
24	1.4187×10^{-4}	1.07734×10^{-3}	-9.35467×10^{-4}
25	6.69402×10^{-6}	1.39615×10^{-3}	-1.38946×10^{-3}
26	6.33693×10^{-6}	1.31578×10^{-3}	-1.30944×10^{-3}
27	5.54163×10^{-5}	4.29335×10^{-4}	-3.73919×10^{-4}
28	5.54163×10^{-5}	4.29335×10^{-4}	-3.73919×10^{-4}
29	4.93773×10^{-5}	3.74964×10^{-4}	-3.25586×10^{-4}
30	4.93773×10^{-5}	3.74964×10^{-4}	-3.25586×10^{-4}
31	1.34192×10^{-5}	1.34192×10^{-5}	-4.21325×10^{-15}

4.3. Computational impact of simplification

To illustrate our method, we performed this analysis on the residual heat removal system. The decomposition, where the sensitivity case is such that z_i remain unchanged and p_i are changed from their upper bounds to their lower bounds can be found in Table 5. We see that most of the change in attributed to the three first probabilities. Of these, the impact of changing the first probability is driven by its individual impact, whereas the impact of changing the second and third probabilities are driven by their interactions with other probabilities. We also see that there are some other probabilities that although have small TOI values, have large interaction affects.

Initially, solving the RHRS case problem instance with narrow intervals for all budget levels took about 30 h. After applying the rules that simplify the reliability function, it could be solved in about 2 h. However, after the simplification, all non-dominated solutions were not found.

A comparison of the simplified method and the exact method can be found in Table 6. In this table, there are the number of potentially ND portfolios and ND portfolios with respect to the simplified reliability function. This ND set corresponds for budget levels 1–5 to the ND set obtained using the original reliability function. However, we see that for budgets 6–13, the average share of ND portfolios is from 39%–85%, whereas for budgets 14–15, the simplified method did not find any ND portfolios.

Table 6
Computations with narrower intervals and simplified reliability function.

Number of improvements	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
#pot. ND portfolios	3	2	5	24	98	27	17	29	39	39	37	25	29	30	35
#ND portfolios (simplified)	3	2	5	7	8	7	6	5	8	6	11	13	6	2	5
Share of actual ND set (%)	100	100	100	100	100	42	75	85	72	74	39	25	67	0	0

5. Discussion

This paper has developed a model for solving the redundancy allocation under interval-valued probabilities. In contrast to the reliability optimization models of [42,43,20] who assume more on system structure, our method is, in principle, applicable to any coherent system. However, for our method to be computationally viable, we need to approximate the reliability as the sum of the minimal cut set probabilities. Also, the degrees of redundancy are assumed to be known for each component, compared to the earlier models.

It should be noted, that the pairwise dominance concept used in this paper is not crucially tied to the use of interval valued probabilities. For instance, when using Bayesian methods, one could define dominance through standard first order stochastic dominance. Hence, although in this paper the computational development is made for interval valued probabilities, similar insights can in principle be obtained using alternative representations of epistemic uncertainty, too.

Based on the computational experiments, our method cannot be used for analyzing the full scale fault tree of a nuclear power plant or perhaps not even its larger subsystems. This observation, however, is true even if there is no interval-uncertainty about the probabilities, because the underlying problem to be solved is a knapsack problem with a nonlinear objective function, which belongs to the class of NP-hard problems. Still, our method can provide insights into how uncertainty impacts the prioritization of risk reduction activities. To ensure computational tractability, one could use other methods such as importance measures to narrow the events that can be included to the reduction portfolio down to a short list of about 10–25 events. Thus, even if some significant events may be missed, at least the events that are on the short list are treated rigorously.

The width of the intervals was more of a factor in this case than the number of improvement actions that could be selected. This could be seen from the results: Narrowing the intervals made it possible to compute all non-dominated portfolios of the resulting problem, while reducing the search space made it possible to solve the problem for only a marginally larger budget.

6. Conclusions

We have presented a method for reliability allocation under interval-valued uncertainty about probabilities. The two-phased algorithm consisting of implicit enumeration and pairwise comparison is capable of solving the resulting the set of non-dominated portfolios that characterize reasonable solutions to this problem. However, the computational complexity increases quickly with the size of the problem. This increase in computational complexity was addressed with state-of-the-art simplification methods. The simplification made it possible to find a significant share of the non-dominated set with considerable less computational effort.

Acknowledgements

Antti Toppila and Ahti Salo acknowledge the financial support of the Nuclear Waste Management Fund (VYR) that was acquired through the Finnish National Nuclear Power Plant Safety Research programme 2011–2014 SAFIR 2014. We also thank Jan-Erik Holmberg and the reviewers for their constructive comments.

References

[1] Rausand M, Høyland A. System reliability theory: models, statistical methods, and applications, vol. 396. New Jersey: John Wiley & Sons; 2004.
[2] Kuo W, Wan R. Recent advances in optimal reliability allocation. In: Computational Intelligence in Reliability Engineering. Springer, Berlin, Heidelberg, 2007. p. 1–36.
[3] Kuo W, Prasad VR. An annotated overview of system-reliability optimization. IEEE Trans. Reliab. 2000;49(2):176–87.
[4] Tillman FA, Hwang C-L, Kuo W. Optimization techniques for system reliability with redundancy? A review. IEEE Trans. Reliab. 1977;26(3):148–55.
[5] Caserta M, Voß S. An exact algorithm for the reliability redundancy allocation problem. Eur J Oper Res 2015;244(1):110–6.
[6] Apostolakis G. The concept of probability in safety assessment of technological systems. Science 1990;250:1359–64.
[7] Walley P. Statistical reasoning with imprecise probabilities, monographs in statistics and applied probability, vol. 42. London: Chapman and Hall; 1991.
[8] Helton J, Johnson J, Oberkampf W. An exploration of alternative approaches to the representation of uncertainty in model predictions. Reliab Eng Syst Saf 2004;85(1–3):39–71.
[9] Augustin T, Coolen FP, de Cooman G, Troffaes MC. Introduction to imprecise probabilities. John Wiley & Sons; 2014.
[10] Limbourg P, Savić R, Petersen J, Kochs H. Modelling uncertainty in fault tree analyses using evidence theory. Proc Inst Mech Eng Part O: J Risk Reliab 2008;222(3):291–302.
[11] Sallak M, Schon W, Aguirre F, et al. Extended component importance measures considering aleatory and epistemic uncertainties. IEEE Trans Reliab 2013;62(1):49–65.
[12] Utkin L, Coolen F. Imprecise reliability: an introductory overview. In: Levitin G, editor. . Intelligence in reliability engineering, Studies in computational intelligence, vol. 40. Berlin/Heidelberg: Springer; 2007. p. 261–306.
[13] Coolen F. On the use of imprecise probabilities on reliability. Quality and Reliability Engineering International 2004;20(3):193–202.
[14] Le Duy T, Vasseur D, Dieulle L, Bérenguer C, Couplet M. Treatment of epistemic uncertainties with non-probabilistic approaches in applications of probabilistic risk assessment. In: Proceedings of PSAM 10 topical meeting. Seattle US, 2010.
[15] Le Duy T, Vasseur D, Couplet M, Dieulle L. Parameter and model uncertainty analysis using Dempster-Shafer theory in nuclear probabilistic risk assessment. In: ANS PSA 2011 international topical meeting on probabilistic safety assessment and analysis, 2011.
[16] Soltani R, Sadjadi S. Reliability optimization through robust redundancy allocation models with choice of component type under fuzziness. Proc Inst Mech Eng Part O: J Risk Reliab 2014:449–59.
[17] Wang W, Xiong J, Xie M. A study of interval analysis for cold-standby system reliability optimization under parameter uncertainty. Comput Ind Eng 2016;97:93–100.
[18] Gupta R, Bhunia A, Roy D. A ga based penalty function technique for solving constrained redundancy allocation problem of series system with interval valued reliability of components. J Comput Appl Math 2009;232(2):275–84.
[19] Bhunia AK, Sahoo L, Roy D. Reliability stochastic optimization for a series system with interval component reliability via genetic algorithm. Appl Math Comput 2010;216(3):929–39.
[20] Feizollahi M, Modarres M. The robust deviation redundancy allocation problem with interval component reliabilities. IEEE Trans Reliab 2012;61(4):957–65.
[21] Liesiö J, Mild P, Salo A. Preference programming for robust portfolio modeling and project selection. Eur J Oper Res 2007;181(3):1488–505.
[22] Liesiö J, Mild P, Salo A. Robust portfolio modeling with incomplete cost information and project interdependencies. Eur J Oper Res 2008;190(3):679–95.
[23] Mild P, Liesiö J. Selecting infrastructure maintenance projects with Robust Portfolio Modeling. Decis Support Syst 2015;77:21–30.
[24] Toppila A, Salo A. Computational framework for prioritization of events in fault tree analysis under interval-valued probabilities. IEEE Trans Reliab 2013;62(3):583–95.
[25] Borgonovo E, Smith CL. A study of interactions in the risk assessment of complex engineering systems: an application to space PSA. Oper Res 2011;59(6):1461–76.
[26] Borgonovo E. The reliability importance of components and prime implicants in coherent and non-coherent systems including total-order interactions. Eur J Oper Res 2010;205(3):485–95.
[27] Borgonovo E. Epistemic uncertainty in the ranking and categorization of probabilistic safety assessment model elements: issues and findings. Risk Anal 2008;28(4):983–1001.
[28] Modarres M. Risk analysis in engineering: techniques, tools, and trends. New York: CRC Press; 2006. p. 223–32.
[29] Zio E. Risk importance measures. In: Pham H, editor. Safety and risk modeling and its applications. London: Springer-Verlag; 2011. p. 151–96.
[30] Baraldi P, Zio E, Compare M. Importance measures in presence of uncertainties.

Reliab Risk Anal: Theory Appl 2009;2(2):30–41.
[31] Weichselberger K. The theory of interval-probability as a unifying concept for uncertainty. Int J Approx Reason 2000;24(2–3):149–70.
[32] Buckley J. Fuzzy probabilities: new approach and applications, vol. 115. Berlin, Heidelberg: Physica-Verlag; 2003.
[33] Dempster A. Upper and lower probabilities induced by a multivalued mapping. Ann Math Stat 1967;38:325–39.
[34] Shafer G. A mathematical theory of evidence, vol. 76. Princeton: Princeton University Press; 1976.
[35] Liesiö J, Mild P, Salo A. Preference programming for robust portfolio modeling and project selection. Eur J Oper Res 2007;181(3):1488–505.
[36] Vilkkumaa E, Salo A, Liesiö J. Multicriteria portfolio modeling for the development of shared action agendas. Group Decis Negot 2014;23(1):49–70.
[37] Liesiö J. Measurable multiattribute value functions for portfolio decision analysis. Decis Anal 2014;11(1):1–20.
[38] Martello S, Toth P. Knapsack problems. New York: Wiley; 1990.
[39] Hoeffding W. On the distribution of the number of successes in independent trials. Ann Math Stat 1956:713–21.
[40] Kuo W, Zhu X. Importance measures in reliability, risk, and optimization: principles and applications. New Delhi: John Wiley & Sons; 2012.
[41] Kuo W, Zhu X. Some recent advances on importance measures in reliability. IEEE Trans Reliab 2012;61(2):344–60.
[42] Sung C, Cho Y. Branch-and-bound redundancy optimization for a series system with multiple-choice constraints. IEEE Trans Reliab 1999;48(2):108–17.
[43] Prasad VR, Kuo W. Reliability optimization of coherent systems. IEEE Trans Reliab 2000;49(3):323–30.